

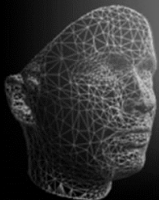
Advanced Topics in Computer Graphics II

Geometry Processing

Differential Forms on Manifolds



December 6, 2024



In the following we distinguish between points and vectors:

Points and vectors

- We call the collection of points $p = (x_1, \dots, x_n)$ with $x_i \in \mathbb{R}, i = 1, \dots, n$, the **manifold** $\mathbb{R}^n$.

- We call the collection of vectors $\begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$ with $v_i \in \mathbb{R}, i = 1, \dots, n$, the **vector space** $\mathbb{R}^n$.

- The functions $v_i : \mathbb{R}^n \rightarrow \mathbb{R}$ that take a point p in the manifold $\mathbb{R}^n$ to its i th-coordinate value are called **coordinate functions**.



Directional derivative

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a real-valued function on the manifold $\mathbb{R}^n$ and let v_p be a vector tangent to manifold $\mathbb{R}^n$, that is, $v_p \in T_p(\mathbb{R}^n)$. The number

$$v_p[f] := \frac{d}{dt} (f(p + tv_p))|_{t=0}$$

is called the directional derivative of f with respect to v_p , if it exists.



Directional derivative

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a real-valued function on the manifold $\mathbb{R}^n$ and let v_p be a vector tangent to manifold $\mathbb{R}^n$, that is, $v_p \in T_p(\mathbb{R}^n)$. The number

$$v_p[f] := \frac{d}{dt} (f(p + tv_p))|_{t=0}$$

is called the directional derivative of f with respect to v_p , if it exists.

- ▶ In the notation $v_p[f]$ the vector v_p "takes in" a function f and gives out a number.
- ▶ The vector v_p is called an operator on the function f .

Note: In the following, we write the components of a vector $v \in T_p(M)$ with upper indices v^i , $i \in 1, \dots, n$:

$$v = \begin{bmatrix} v^1 \\ \vdots \\ v^n \end{bmatrix}$$

Furthermore, we write components of a point $p = (x^i)$ like this as well.



Example: Computation of $v_p[f]$

Given a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and a vector $v_p = \begin{bmatrix} v^1 \\ v^2 \\ v^3 \end{bmatrix}_p \in T_p(\mathbb{R}^3)$ at the point $p = (p^1, p^2, p^3)$ we compute $v_p[f]$ as follows:

$$\begin{aligned} v_p[f] &= \frac{d}{dt} (f(p + tv_p))|_{t=0} \\ &= \frac{d}{dt} f(\underbrace{p^1 + tv^1}_{x^1(t)}, \underbrace{p^2 + tv^2}_{x^2(t)}, \underbrace{p^3 + tv^3}_{x^3(t)})|_{t=0} \\ &= \frac{\partial f}{\partial x^1} \Big|_{x^1(0)} \cdot \frac{dx^1}{dt} \Big|_{t=0} + \frac{\partial f}{\partial x^2} \Big|_{x^2(0)} \cdot \frac{dx^2}{dt} \Big|_{t=0} + \frac{\partial f}{\partial x^3} \Big|_{x^3(0)} \cdot \frac{dx^3}{dt} \Big|_{t=0} \\ &= \frac{\partial f}{\partial x^1} \Big|_{x^1(0)} \cdot v^1 + \frac{\partial f}{\partial x^2} \Big|_{x^2(0)} \cdot v^2 + \frac{\partial f}{\partial x^3} \Big|_{x^3(0)} \cdot v^3 \\ &= \sum_{i=1}^3 v^i \cdot \frac{\partial f}{\partial x^i} \Big|_{t=0} \end{aligned}$$



Unit vectors and directional derivatives

Using $v_p[f] = \sum_{i=1}^3 v^i \cdot \frac{\partial f}{\partial x^i} \big|_{t=0}$ it is easy to compute how the standard Euclidean

vectors unit vectors $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}_p$, $e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}_p$, $e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}_p$ act on f :

$$e_1[f] = \frac{\partial}{\partial x^1}(f), e_2[f] = \frac{\partial}{\partial x^2}(f), e_3[f] = \frac{\partial}{\partial x^3}(f)$$

Identification of vectors and partial differential operators

The Euclidian vectors e_i can be identified with the partial differential operators $\frac{\partial}{\partial x^i}$.

Furthermore, each vector $v_p \in T_p(\mathbb{R}^3)$ can be identified with the differential operator

$$v_p = \begin{bmatrix} v^1 \\ v^2 \\ v^3 \end{bmatrix}_p = v^1 \frac{\partial}{\partial x^1} \big|_p + v^2 \frac{\partial}{\partial x^2} \big|_p + v^3 \frac{\partial}{\partial x^3} \big|_p.$$



Einstein summing notation

Einstein summation is a shorthand notation for summing over repeated upper and lower indices:

$$x^i y_i := \sum_{i=1}^n x^i y_i \quad x_i y^i := \sum_{i=1}^n x_i y^i$$

Note on a simplified way of writing expressions A look at the equations of this paragraph show that there is always a summation over indices which occur twice, and only for twice-repeated indices. It is therefore possible, without detracting from clarity, to omit the sum sign. For this we introduce a rule: if an index in an expression appears twice, then a sum is implicitly taken over this index, unless specifically noted to the contrary.¹

Example

$$T_{lm}^{ijk} \Lambda_{jk}^l := \sum_j \sum_k \sum_l T_{lm}^{ijk} \Lambda_{jk}^l$$

¹Einstein, “Die Grundlage der allgemeinen Relativitätstheorie” (1916)



To exploit the Einstein summing notation elements of the vector space

$V = \text{span}\{e_1, e_2, e_3\}$ are written using **vector coefficients** written with **upper indices** as

$$v = v^1 e_1 + v^2 e_2 + v^3 e_3 = \begin{pmatrix} v^1 \\ v^2 \\ v^3 \end{pmatrix} = v^i e_i$$

Note that for a valid summation, if an index appears twice, it has to appear once as an upper and once as a lower index.

For fractions, indices that are used in the denominator are read as the opposite position. Therefore,

$$v = v^1 \frac{\partial}{\partial x^1} + v^2 \frac{\partial}{\partial x^2} + v^3 \frac{\partial}{\partial x^3} = \begin{pmatrix} v^1 \\ v^2 \\ v^3 \end{pmatrix} = v^i \frac{\partial}{\partial x^i}$$

still adheres to the convention as the i in the denominator is interpreted as a lower index. We sometimes also write $\frac{\partial}{\partial x^i} = \partial x_i$.



Let $\langle \mathbf{v}, \mathbf{w} \rangle : T_p(M) \times T_p(M) \rightarrow \mathbb{R}$ be an inner product on every vector space $T_p(M)$ of the manifold M .

Metric tensor

The inner product of the basis elements is expressed by an $n \times n$ matrix G called **metric tensor** (which is symmetric by the definition of the inner product) with entries

$$g_{ij} = \left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle$$

so that

$$\langle \mathbf{v}, \mathbf{w} \rangle_G = \sum_i \sum_j v^i w^j g_{ij} = v^i w^j g_{ij}$$

for any two vectors $\mathbf{v}, \mathbf{w} \in T_p(M)$.

This can be written in matrix form

$$\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^t \mathbf{G} \mathbf{w}$$

where $\mathbf{v} = [v^1, \dots, v^n]^t$, $\mathbf{w} = [w^1, \dots, w^n]^t$, and $[g_{ij}] = \mathbf{G}$

Under an Euclidean (**flat**) metric and Cartesian coordinates the metric tensor is the identity matrix $\mathbf{I}$ and therefore the inner product of two vectors is equivalent to the dot product of their components.



Directional Derivatives

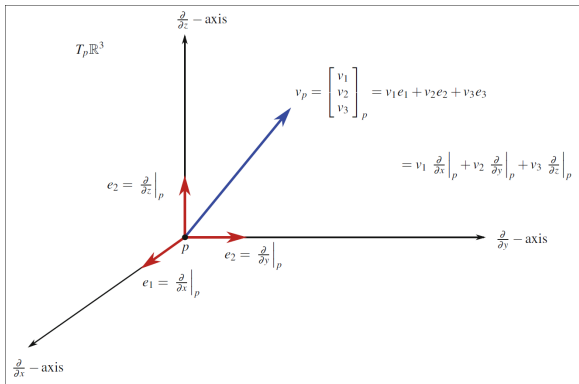


Figure: Identifying the euclidean vectors e_i with the partial differential operators $\frac{\partial}{\partial x^i}$ (Figure from²)

²J. P. Fortney. *A Visual Introduction to Differential Forms and Calculus on Manifolds*. en. Cham: Springer International Publishing, 2018. URL: <http://link.springer.com/10.1007/978-3-319-96992-3> (visited on 11/03/2021).



Definition: Differential One-Form

A differential one-form α on the manifold $\mathbb{R}^n$ is a linear functional on the set of tangent vectors to the manifold $T_p(\mathbb{R}^n)$, i.e. $\alpha : T_p(\mathbb{R}^n) \rightarrow \mathbb{R}$ with

$$\begin{aligned}\alpha(v_p + w_p) &= \alpha(v_p) + \alpha(w_p), \\ \alpha(av_p) &= a\alpha(v_p)\end{aligned}$$

$\forall v_p, w_p \in T_p(\mathbb{R}^n)$ and $a \in \mathbb{R}$.

Comments

- ▶ The dual space of $T_p(\mathbb{R}^n)$ is denoted as $T_p^*(\mathbb{R}^n)$. It is called **cotangent space to $\mathbb{R}^n$** . Its elements are called **cotangent vectors**.
- ▶ The basis vectors (one-forms) in the cotangent space $T_p^*(\mathbb{R}^n)$ are denoted as $dx^1, \dots, dx^n$.
- ▶ From the definition of a dual basis we know how these one-forms act on the basis elements of the tangent space $T_p(\mathbb{R}^n)$:

$$dx^i \left(\frac{\partial}{\partial x^j} \right) = \delta_j^i = \begin{cases} 1, & i = j \\ 0, & \text{else} \end{cases}$$



Remarks:

- Each basis element $dx^i \in T_p^*(\mathbb{R}^n)$, $i = 1, \dots, n$ finds the projection of v to the appropriate axis:

$$dx^i(v_p) = dx^i \begin{bmatrix} v_p^1 \\ \vdots \\ v_p^n \end{bmatrix} = v_p^i.$$

- We can view dx^i as finding the projection of the vector v_p onto the $\frac{\partial}{\partial x^i} \big|_p$ coordinate axis of $T_p(\mathbb{R}^n)$.

To keep in line with Einstein summation notation from now on we will write the **dual basis elements**, with **upper indices** as e^1, e^2, e^3 and the **co-vector coefficients** using **lower indices**:

$$\alpha = \alpha_1 e^1 + \alpha_2 e^2 + \alpha_3 e^3 = \begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 \end{pmatrix} = \alpha_i e^i$$



Definition: Differential df of f

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function on the manifold $\mathbb{R}^n$. The differential df of f is defined to be the one-form on $\mathbb{R}^n$ such that for all vectors v_p we have

$$df(v_p) = v_p[f]$$

Since $df(v_p) = \sum_{i=1}^n \frac{\partial f}{\partial x^i} \cdot v_p^i$ and $v_p^i = dx^i(v_p)$ we get

$$df(v_p) = \sum_{i=1}^n \frac{\partial f}{\partial x^i} \cdot dx^i(v_p) = \left(\sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i \right)(v_p)$$

and so we have

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i$$



Since the inner product $\langle v, w \rangle_G$ of two vectors in $T_p(M)$ is a linear functional of w when v is held fixed, the function α defined by

$$\alpha(w) = \langle v, w \rangle_G$$

is a linear functional, i.e. a one-form on $T_p(M)$. It can be represented by the standard dual basis, $\alpha = \sum_j \alpha_j dx^j$.

Using the mapping between forms and vectors, we see that

$$\begin{aligned} \alpha &= \sum_j \alpha_j dx^j = \sum_j \alpha(e_j) dx^j = \sum_j \langle v, e_j \rangle_G dx^j \\ &= \sum_j \left\langle \sum_i (v^i e_i), e_j \right\rangle_G dx^j = \sum_j \underbrace{\left(\sum_i v^i g_{ij} \right)}_{\alpha_j} dx^j \end{aligned}$$



Using

$$\alpha_j = \sum_i v^i g_{ij}$$

and taking into account that the inner product is defined to be non-degenerate, the inverse matrix $G^{-1} = (g^{ij})$ exists and is symmetric. Therefore we can compute

$$v = G^{-1}\alpha$$

Notice, that in the special case of a Euclidean metric where $G = I$ we get $\alpha_j = v^j$.

G^{-1} provides an inner product on $T_p^*(M)$ that is equivalent to the original inner product on $T_p(M)$: Let $\alpha = v^t G$ and $\beta = w^t G$. Then

$$\langle \alpha, \beta \rangle_{G^{-1}} = \alpha G^{-1} \beta = v^t G G^{-1} G^t w = v^t G^t w = v^t G w.$$

Dual metric tensor

Let M be a n -dimensional manifold. Then G^{-1} provides an inner product on $T_p^*(M)$ which for two one-forms α, β is given by

$$\langle \alpha, \beta \rangle_{G^{-1}} = \alpha G^{-1} \beta^t$$

where $\alpha = [\alpha_1, \dots, \alpha_n], \beta = [\beta_1, \dots, \beta_n]$



Musical isomorphisms



Let $M = \mathbb{R}^3$. At any point $p \in M$ the tangent and cotangent space are defined by:

$$\begin{aligned} T_p \mathbb{R}^3 &= \text{span} \left\{ \left. \frac{\partial}{\partial x^1} \right|_p, \left. \frac{\partial}{\partial x^2} \right|_p, \left. \frac{\partial}{\partial x^3} \right|_p \right\} \\ &= \text{span} \left\{ \frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3} \right\} \end{aligned}$$

$$\begin{aligned} T_p^* \mathbb{R}^3 &= \text{span} \left\{ dx^1|_p, dx^2|_p, dx^3|_p \right\} \\ &= \text{span} \{ dx^1, dx^2, dx^3 \} \end{aligned}$$

The $\flat$ isomorphism

$$\flat : T_p M \rightarrow T_p^* M$$

$$v^i \frac{\partial}{\partial x^i} \mapsto \hat{v}_i dx^i \text{ where } \hat{v}_i = g_{ij} v^j.$$

The $\sharp$ isomorphism

$$\sharp : T_p^* M \rightarrow T_p M$$

$$\alpha_i dx^i \mapsto \hat{\alpha}^i \frac{\partial}{\partial x^i} \text{ where } \hat{\alpha}^i = g^{ij} \alpha_j.$$

The notation using the “musical isomorphisms” can be explained by Einstein summation, as flat *raises* the pitch and sharp *lowers* the pitch, i.e., the indices of the coefficients.



Example 1:

In an Euclidean (flat) space, the metric tensor is the identity and it holds

$$v^{\flat} = \left(v^i \frac{\partial}{\partial x^i} \right)^{\flat} = v_i dx^i$$
$$\alpha^{\sharp} = \left(\alpha_i dx^i \right)^{\sharp} = \alpha^i \frac{\partial}{\partial x^i}$$

Example 2:

Suppose $\alpha(v) = \langle u, v \rangle_G$ for all $v \in V$. Then

$$\alpha = \alpha_1 e^1 + \cdots + \alpha_n e^n \Rightarrow^{\sharp} u = u^1 e_1 + \cdots + u^n e_n$$
$$u = u^1 e_1 + \cdots + u^n e_n \Rightarrow^{\flat} \alpha = \alpha_1 e^1 + \cdots + \alpha_n e^n$$



Wedge product of two one-forms

The wedge product of two one-forms dx^i and dx^j is defined by

$$dx^i \wedge dx^j(v_p, w_p) := \begin{vmatrix} dx^i(v_p) & dx^i(w_p) \\ dx^j(v_p) & dx^j(w_p) \end{vmatrix}$$

Note, that is the volume of the parallelepiped spanned by $\begin{bmatrix} dx^i(v_p) \\ dx^j(v_p) \end{bmatrix}$ and $\begin{bmatrix} dx^i(w_p) \\ dx^j(w_p) \end{bmatrix}$.



The Wedge Product

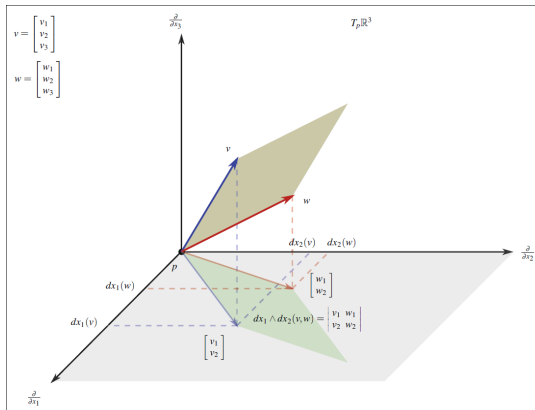


Figure: The parallelepiped spanned by v and w (brown) is projected onto the $\partial x_1 \partial x_2$ -plane in $T_p \mathbb{R}^3$ (green). The wedge product $dx^1 \wedge dx^2$ finds the volume of this projected area (Figure from³)

³J. P. Fortney. *A Visual Introduction to Differential Forms and Calculus on Manifolds*. en. Cham: Springer International Publishing, 2018. URL: <http://link.springer.com/10.1007/978-3-319-96992-3> (visited on 11/03/2021).



Two-forms

- ▶ The wedge product of two one-forms is called two-form.
- ▶ The vector space of two-forms at p is denoted as $\Lambda_p^2(\mathbb{R}^n)$. Keeping in line with this convention, $T_p^*(\mathbb{R}^n)$ is written as $\Lambda_p^1(\mathbb{R}^n)$.

Properties of the wedge product

- ▶ $dx^i \wedge dx^j = -dx^j \wedge dx^i$ (skew symmetry).
- ▶ $dx^i \wedge dx^i = 0$, $i = 1, \dots, n$
- ▶ $\{dx^1 \wedge dx^2, dx^2 \wedge dx^3, dx^3 \wedge dx^1\}$ is a basis of $\Lambda_p^2(\mathbb{R}^3)$,
 $\{dx^1 \wedge dx^2, dx^1 \wedge dx^3, dx^1 \wedge dx^4, dx^2 \wedge dx^3, dx^2 \wedge dx^4, dx^3 \wedge dx^4\}$ is a basis of $\Lambda_p^2(\mathbb{R}^4)$, etc.
- ▶ The basis of $\Lambda_p^2(\mathbb{R}^n)$ contains $\frac{(n)(n-1)}{2}$ elements.



The Wedge product

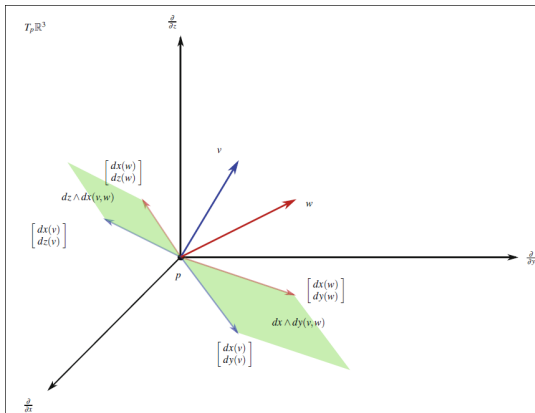


Figure: The action of the two-form $dx \wedge dy + dz \wedge dx$ on the two vectors v and w . The two form $dx \wedge dy + dz \wedge dx$ finds the sum of the areas of the two projected parallelepipeds (Figure from⁴)

⁴J. P. Fortney. *A Visual Introduction to Differential Forms and Calculus on Manifolds*. en. Cham: Springer International Publishing, 2018. URL: <http://link.springer.com/10.1007/978-3-319-96992-3> (visited on 11/03/2021).



Definition: Wedge product of n one-forms

Let $v_1, \dots, v_n$ represent n vectors, we define the wedge product of n one-forms as

$$dx^{i_1} \wedge dx^{i_2} \wedge \dots \wedge dx^{i_n}(v_1, \dots, v_n) := \begin{vmatrix} dx^{i_1}(v_1) & dx^{i_1}(v_2) & \dots & dx^{i_1}(v_n) \\ dx^{i_2}(v_1) & dx^{i_2}(v_2) & \dots & dx^{i_2}(v_n) \\ \vdots & \vdots & \ddots & \vdots \\ dx^{i_n}(v_1) & dx^{i_n}(v_2) & \dots & dx^{i_n}(v_n) \end{vmatrix}$$

If we have any two of the one-forms the same, that is, $i_j = i_k$ for some $j \neq k$ then we have two rows that are the same, which gives a value of zero.

This determinant is the signed volume of the parallelepiped spanned by the vectors

$$[dx^{i_1}(v_j) dx^{i_2}(v_j) \dots dx^{i_n}(v_j)]^t, j = 1, \dots, n.$$

Definition n -form

An n -form is defined to be a linear combination of elements of the form $dx^{i_1} \wedge dx^{i_2} \wedge \dots \wedge dx^{i_n}$ and the space of n -forms on M at point p is denoted by $\Lambda_p^n(M)$.

We can calculate the volume of a m -parallelepiped using the matrix A with its m base vectors as columns as $\text{vol}(P)^2 = \det(A^T A)$.

Theorem

Let P be an m -dimensional parallelepiped spanned by the m vectors $a_1, \dots, a_m \in \mathbb{R}^n$

and $A = \left(\begin{array}{c|c|c|c} a_1 & a_2 & \dots & a_m \end{array} \right)$. Then

$$\text{vol}(P)^2 = \det(A^T A)$$

Corollary

If $m = n$ then

$$\text{vol}(P) = |\det(A)|$$

Note that $\det(A^T A) = \det(A)^2$.

Proof.

Proof by induction over m . For $m = 1$, we have $A^T A = a_1^T a_1 = \|a_1\|^2$.

Now let $m > 1$ and assume that the proposition holds for $m - 1$. Let

$$A = \left(\begin{array}{c|c|c|c} a_1 & a_2 & \dots & a_m \end{array} \right) \text{ and } D = \left(\begin{array}{c|c|c|c} a_1 & a_2 & \dots & a_{m-1} \end{array} \right). \text{ Then we have}$$

$$A = \left(\begin{array}{c|c} D & a_m \end{array} \right) \text{ and can decompose } a_m \text{ into } a_m = a_m^\perp + a_m^\parallel, \text{ where}$$

$$a_m^\parallel \in \underbrace{\text{span}(a_1, \dots, a_{m-1})}_{\mathbb{R}^n \times \dots \times \mathbb{R}^n \text{ (m-1 times)}}, \text{ and } a_m^\perp \in \underbrace{\text{span}(a_1, \dots, a_{m-1})^\perp}_{\mathbb{R}^n}, \text{ i.e.,}$$

$$a_m = a_m^\perp + \sum_{i=1}^{m-1} \alpha_i a_i.$$

$$\text{Let } \hat{A} = \left(\begin{array}{c|c|c|c} a_1 & a_2 & \dots & a_m^\perp \end{array} \right).$$

□

Proof.

Then

$$A = \left(\begin{array}{c|c|c|c} a_1 & a_2 & \dots & a_m \end{array} \right) = \left(\begin{array}{c|c|c|c} a_1 & a_2 & \dots & a_m^\perp \end{array} \right) \underbrace{\left(\begin{array}{c|c|c|c} 1 & 0 & 0 & \alpha_1 \\ 0 & \ddots & 0 & \vdots \\ 0 & 0 & 1 & \alpha_{m-1} \\ 0 & 0 & 0 & 1 \end{array} \right)}_E$$

Therefore we have, $\det(A^T A) = \det(E^T \hat{A}^T \hat{A} E) = \det(\hat{A}^T \hat{A})$ and with

$$\hat{A}^T \hat{A} = \left(\begin{array}{c} D^T \\ (a_m^\perp)^T \end{array} \right) \left(\begin{array}{c|c} D & a_m^\perp \end{array} \right) = \left(\begin{array}{c|c} D^T D & D^T a_m^\perp \\ (a_m^\perp)^T D & a_m^\perp a_m \end{array} \right) = \left(\begin{array}{c|c} D^T D & 0 \\ 0 & a_m^\perp a_m \end{array} \right)$$

we get

$$\det(\hat{A}^T \hat{A}) = \det(D^T D) \cdot a_m^\perp a_m = \text{vol}(a_1, \dots, a_{m-1})^2 \cdot \|a_m^\perp\|^2 = \text{vol}(a_1, \dots, a_m)^2$$

□

Example: Basis for $\Lambda^2(\mathbb{R}^3)$, $\Lambda^3(\mathbb{R}^4)$, $\Lambda^k(\mathbb{R}^n)$

$$\Lambda^2(\mathbb{R}^3) : \{dx^1 \wedge dx^2, dx^2 \wedge dx^3, dx^3 \wedge dx^1\}$$

$$\Lambda^3(\mathbb{R}^4) : \{dx^1 \wedge dx^2 \wedge dx^3, dx^1 \wedge dx^2 \wedge dx^4, dx^1 \wedge dx^3 \wedge dx^4, dx^2 \wedge dx^3 \wedge dx^4\}$$

$$\Lambda^k(\mathbb{R}^n) : \{dx^I : I \in \mathcal{J}_{k,n} = \{(i_1, i_2, \dots, i_k) : 1 \leq i_1 < i_2 < \dots < i_k \leq n\}\}$$

Example

For $k = 2$ and $n = 4$ we have $I \in \mathcal{J}_{2,4} = \{12, 13, 14, 23, 24, 34\}$

Wedge product of two arbitrary forms

Let $\alpha \in \Lambda^k(\mathbb{R})^n$ and $\beta \in \Lambda^l(\mathbb{R})^n$ where $\alpha = \sum_I a_I dx^I$ and $\beta = \sum_J b_J dx^J$. Then

$$\alpha \wedge \beta = \left(\alpha = \sum_I a_I dx^I \right) \wedge \left(\beta = \sum_J b_J dx^J \right) = \sum_{I,J} a_I b_J dx^I \wedge dx^J$$

In the sum on the right hand side we have $dx^I \wedge dx^J = 0$ if I and J have any indices in common.

If I and J are disjoint the $dx^I \wedge dx^J = \pm dx^K$ where $K = I \cup J$, but is reordered to be in increasing order and elements with repeated indices are dropped.

Wedge Product of two arbitrary forms

$$\alpha \wedge \beta = \left(\alpha = \sum_I a_I dx^I \right) \wedge \left(\beta = \sum_J b_J dx^J \right) = \sum_K \left(\sum_{I \cup J = K} \pm a_I b_J \right) dx^K$$

Exterior Algebra

The vector space

$$\Lambda(\mathbb{R}^n) := \bigoplus_{k=0}^n \Lambda^k(\mathbb{R}^n)$$

the is called the exterior algebra of $\mathbb{R}^n$. Since each $\Lambda^k(\mathbb{R}^n)$ has the dimension $\binom{n}{k}$ we deduce that $\dim(\Lambda(\mathbb{R}^n)) = 2^n$.

Algebraic properties

Let $\omega, \omega_1, \omega_2$ be k -forms, η, η_1, η_2 are l -forms and θ is a m -form. Then the following properties hold:

1. $a\omega \wedge \eta = \omega \wedge a\eta = a(\omega \wedge \eta)$, $a \in \mathbb{R}$ (scalar multiplication)
2. $(\omega \wedge \eta) \wedge \theta = \omega \wedge (\eta \wedge \theta)$ (Associativity)
3. $(\omega_1 + \omega_2) \wedge (\eta) = (\omega_1 \wedge \eta) + (\omega_2 \wedge \eta)$ and $\omega \wedge (\eta_1 + \eta_2) = \omega \wedge \eta_1 + \omega \wedge \eta_2$ (Distributivity)
4. $\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega$ (Anticommutativity)
5. $\omega \wedge \omega = 0$ (follows from 4)



Theorem and definition: Exterior derivative

There exists a unique operator

$$d : \Lambda^n(M) \rightarrow \Lambda^{n+1}(M)$$

called **exterior derivative** that satisfy the following four properties. Suppose α, β are n -forms and ω is an m -form on M , the operator d satisfies

1. $d(\alpha + \beta) = d\alpha + d\beta$,
2. $d(\alpha \wedge \omega) = d\alpha \wedge \omega + (-1)^n \alpha \wedge d\omega$ (Leibniz's Law or the chain rule),
3. $d(d\alpha) = 0$,
4. for each function f on M , $df = \sum \frac{\partial f}{\partial x^i} dx^i$.

The operator d , when applied to a zero-form function is its directional derivative.



Exterior derivative of an n -form

$$d\left(\sum \alpha_{i_1 \dots i_n} dx^{i_1} \wedge \dots \wedge dx^{i_n}\right) = \sum_{j=1}^n \sum \frac{\partial \alpha_{i_1 \dots i_n}}{\partial x^{i_j}} dx^{i_j} \wedge dx^{i_1} \wedge \dots \wedge dx^{i_n}.$$

Geometric nature of the exterior derivative

The exterior derivative measures the variation of a p -form simultaneously in each of the p directions of a p -dimensional parallelepiped, and is therefore the natural generalization of the one-dimensional differential operator $\frac{d}{dt}$.



Example

Given two functions $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then we get

$$\begin{aligned} df \wedge dg &= \left(\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \right) \wedge \left(\frac{\partial g}{\partial x} dx + \frac{\partial g}{\partial y} dy \right) \\ &= \frac{\partial f}{\partial x} dx \wedge \left(\frac{\partial g}{\partial x} dx + \frac{\partial g}{\partial y} dy \right) + \frac{\partial f}{\partial y} dy \wedge \left(\frac{\partial g}{\partial x} dx + \frac{\partial g}{\partial y} dy \right) \\ &= \left(\frac{\partial f}{\partial x} \frac{\partial g}{\partial y} dx \wedge dy - \frac{\partial f}{\partial y} \frac{\partial g}{\partial x} dx \wedge dy \right) \\ &= \underbrace{\begin{vmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{vmatrix}}_{\text{Jacobian}} \underbrace{dx \wedge dy}_{\text{area form}} \end{aligned}$$

A similar result holds for functions $f, g : \mathbb{R}^3 \rightarrow \mathbb{R}$



Example

Let $x = r\cos(\theta)$, $y = r\sin(\theta)$. Then

$$\begin{aligned} dx \wedge dy &= \begin{vmatrix} \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial r} \\ \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial r} \end{vmatrix} d\theta \wedge dr \\ &= \begin{vmatrix} \frac{\partial(r\cos(\theta))}{\partial \theta} & \frac{\partial(r\cos(\theta))}{\partial r} \\ \frac{\partial(r\sin(\theta))}{\partial \theta} & \frac{\partial(r\sin(\theta))}{\partial r} \end{vmatrix} d\theta \wedge dr \\ &= \begin{vmatrix} -r\sin(\theta) & \cos(\theta) \\ r\cos(\theta) & \sin(\theta) \end{vmatrix} d\theta \wedge dr \\ &= -r d\theta \wedge dr \end{aligned}$$



Inner product on Λ_p^k

On the vector spaces $\Lambda_p^k, 0 \leq k \leq n$ an inner product denoted by $\langle \cdot, \cdot \rangle_\wedge$ is defined by

$$\langle \alpha_1 \wedge \cdots \wedge \alpha_k, \beta_1 \wedge \cdots \wedge \beta_k \rangle_\wedge := \det(\langle \alpha_i, \beta_j \rangle_{G^{-1}}),$$

for all $\alpha_i, \beta_i \in \mathbb{R}^n$ and extending $\langle \cdot, \cdot \rangle_\wedge$ by bilinearity.

Note that if $(\sigma^1, \dots, \sigma^n)$ is an orthonormal basis of $T_p^*(\mathbb{R}^n)$, then the basis of $\Lambda_k(\mathbb{R}^n)$ consisting of the σ_I , where $I = \{i_1, \dots, i_k\}$, with $1 \leq i_1 < \dots < i_k \leq n$ is an orthonormal basis of $\Lambda^k(\mathbb{R}^n)$.



Example

$\Lambda_p^1(\mathbb{R}^3) = \text{span}\{dx^1, dx^2, dx^3\}$ where dx^1, dx^2 , and dx^3 are orthonormal.

$$\begin{aligned}\langle \alpha, \beta \rangle_{\wedge} &= \left\langle \sum_{i=1}^3 \alpha_i dx^i, \sum_{i=1}^3 \beta_i dx^i \right\rangle_{\wedge} \\&= \sum_{i=1}^3 \sum_{j=1}^3 \alpha_i \beta_j \langle dx^i, dx^j \rangle_{\wedge} \\&= [\alpha_1, \alpha_2, \alpha_3] \begin{pmatrix} \langle dx^1, dx^1 \rangle & \langle dx^1, dx^2 \rangle & \langle dx^1, dx^3 \rangle \\ \langle dx^2, dx^1 \rangle & \langle dx^2, dx^2 \rangle & \langle dx^2, dx^3 \rangle \\ \langle dx^3, dx^1 \rangle & \langle dx^3, dx^2 \rangle & \langle dx^3, dx^3 \rangle \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \\&= [\alpha_1, \alpha_2, \alpha_3] \mathbf{G}^{-1} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \\&= [\alpha_1, \alpha_2, \alpha_3] \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}\end{aligned}$$



Example

$$\Lambda_p^2(\mathbb{R}^3) = \text{span}\{\underbrace{dx^1 \wedge dx^2}_{\sigma^1}, \underbrace{dx^1 \wedge dx^3}_{\sigma^2}, \underbrace{dx^2 \wedge dx^3}_{\sigma^3}\}, \text{ orthonormal}$$

$$\begin{aligned}\langle \alpha, \beta \rangle_{\wedge} &= \left\langle \sum_{i=1}^3 \alpha_i \sigma^i, \sum_{i=1}^3 \beta_i \sigma^i \right\rangle_{\wedge} \\ &= \sum_{i=1}^3 \sum_{j=1}^3 \alpha_i \beta_j \langle \sigma^i, \sigma^j \rangle_{\wedge} \\ &= [\alpha_1, \alpha_2, \alpha_3] \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}\end{aligned}$$



Orientation of a vector space

An **orientation** of a vector space V of dimension n is given by the choice of some basis $(e_1, \dots, e_n)$. We say that a basis $(u_1, \dots, u_n)$ of V is **positively oriented** iff $\det(u_1, \dots, u_n) > 0$, otherwise it is **negatively oriented**. An **oriented vector space** is a vector space V together with an orientation of V .

Notice, if V is oriented by the basis $(e_1, \dots, e_n)$ and σ is any permutation of $\{1, \dots, n\}$ then the basis $(e_{\sigma(1)}, \dots, e_{\sigma(n)})$ has positive orientation iff the signature $\text{sgn}(\sigma)$ of the permutation σ is even.



Proposition

Let $T_p^*(\mathbb{R}^n)$ be oriented by the **orthonormal** basis $\{dx^1, \dots, dx^n\}$. For each k -form $\alpha \in \Lambda_p^k(\mathbb{R}^n)$ there is a unique $(n-k)$ -form $\star\alpha \in \Lambda_p^{n-k}(\mathbb{R}^n)$ such that for any $(n-k)$ -form $\beta \in \Lambda_p^{n-k}(\mathbb{R}^n)$ we have

$$\alpha \wedge \beta = \langle \star\alpha, \beta \rangle_{\wedge} dx^1 \wedge \dots \wedge dx^n.$$

Proof.

Since $\Lambda_p^n(\mathbb{R}^n)$ has dimension 1, $\{dx^1 \wedge \dots \wedge dx^n\}$ is a basis of Λ_p^n . It follows that for any fixed $\alpha \in \Lambda_p^k(\mathbb{R}^n)$, the linear map $\lambda_\alpha : \Lambda_p^{n-k}(\mathbb{R}^n) \rightarrow \Lambda_p^n(\mathbb{R}^n)$ given by

$$\lambda_\alpha(\beta) = \alpha \wedge \beta$$

is of the form

$$\lambda_\alpha(\beta) = f_\alpha(\beta) dx^1 \wedge \dots \wedge dx^n$$

for some linear form $f_\alpha : \Lambda_p^{n-k}(\mathbb{R}^n) \rightarrow \mathbb{R}$. But then, by the duality induced by the inner product $\langle \cdot, \cdot \rangle_{\wedge}$ on $\Lambda_p^{n-k}(\mathbb{R}^n)$, there is a unique vector $\star\alpha \in \Lambda_p^k(\mathbb{R}^n)$ such that

$$\alpha \wedge \beta = \lambda_\alpha(\beta) = f_\alpha(\beta) dx^1 \wedge \dots \wedge dx^n = \langle \star\alpha, \beta \rangle_{\wedge} dx^1 \wedge \dots \wedge dx^n$$





Definition Hodge star

The linear operator $\star : \Lambda_p^k(\mathbb{R}^n) \rightarrow \Lambda_p^{n-k}(\mathbb{R}^n)$ defined by the above proposition is called the Hodge $\star$ -operator.

We can unpack this implicit definition as follows:

$$\underbrace{\alpha}_{k\text{-form}} \wedge \underbrace{\beta}_{(n-k)\text{-form}} = \underbrace{\langle \underbrace{\star\alpha}_{(n-k)\text{-form}}, \underbrace{\beta}_{(n-k)\text{-form}} \rangle}_{\text{a number}} \wedge \underbrace{dx^1 \wedge \dots \wedge dx^n}_{n\text{-form (volume form)}}$$

Note that this formulation holds for **orthonormal** basis vectors (i.e., $G = I$). In general the right-hand side can be replaced by the volume-form

$$\alpha \wedge \beta = \langle \star\alpha, \beta \rangle \wedge dA$$

where $dA = \sqrt{\det(\langle \sigma^j, \sigma^j \rangle_{G^{-1}})} \sigma^1 \wedge \dots \wedge \sigma^n$ for an arbitrary basis $\{\sigma^1, \dots, \sigma^n\}$.



Example $\star : \Lambda_p^1(\mathbb{R}^3) \rightarrow \Lambda_p^{3-1}(\mathbb{R}^3)$: Given $dx^1 \in \Lambda_p^1\mathbb{R}^3$. How to get $\star dx^1 \in \Lambda_p^2(\mathbb{R}^3)$?
 Write $\star dx^1 = \alpha_1 dx^1 \wedge dx^2 + \alpha_2 dx^1 \wedge dx^3 + \alpha_3 dx^2 \wedge dx^3$ for some unknown $\alpha_1, \alpha_2, \alpha_3$
 and let $\beta = \beta_1 dx^1 \wedge dx^2 + \beta_2 dx^1 \wedge dx^3 + \beta_3 dx^2 \wedge dx^3 \in \Lambda_p^{3-1}(\mathbb{R}^3)$ be any two-form
 written in the same basis. Then we have

$$\begin{aligned}
 dx^1 \wedge \beta &= dx^1 \wedge (\beta_1 dx^1 \wedge dx^2 + \beta_2 dx^1 \wedge dx^3 + \beta_3 dx^2 \wedge dx^3) \\
 &= \beta_3 dx^1 \wedge dx^2 \wedge dx^3 \\
 &\stackrel{\text{def}}{=} \langle \star dx^1, \beta \rangle_{\wedge} dx^1 \wedge dx^2 \wedge dx^3 \\
 &= \left\langle \left(\alpha_1 dx^1 \wedge dx^2 + \alpha_2 dx^1 \wedge dx^3 + \alpha_3 dx^2 \wedge dx^3 \right), \right. \\
 &\quad \left. \left(\beta_1 dx^1 \wedge dx^2 + \beta_2 dx^1 \wedge dx^3 + \beta_3 dx^2 \wedge dx^3 \right) \right\rangle_{\wedge} dx^1 \wedge dx^2 \wedge dx^3 \\
 &= [\alpha_1, \alpha_2, \alpha_3] \\
 &\quad \left(\begin{array}{ccc} \langle dx^1 \wedge dx^2, dx^1 \wedge dx^2 \rangle_{\wedge} & \langle dx^1 \wedge dx^2, dx^1 \wedge dx^3 \rangle_{\wedge} & \langle dx^1 \wedge dx^2, dx^2 \wedge dx^3 \rangle_{\wedge} \\ \langle dx^1 \wedge dx^3, dx^1 \wedge dx^2 \rangle_{\wedge} & \langle dx^1 \wedge dx^3, dx^1 \wedge dx^3 \rangle_{\wedge} & \langle dx^1 \wedge dx^3, dx^2 \wedge dx^3 \rangle_{\wedge} \\ \langle dx^2 \wedge dx^3, dx^1 \wedge dx^2 \rangle_{\wedge} & \langle dx^2 \wedge dx^3, dx^1 \wedge dx^3 \rangle_{\wedge} & \langle dx^2 \wedge dx^3, dx^2 \wedge dx^3 \rangle_{\wedge} \end{array} \right) \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \\
 &= \alpha_1 \beta_1 \langle dx^1 \wedge dx^2, dx^1 \wedge dx^2 \rangle_{\wedge} + \alpha_2 \beta_2 \langle dx^1 \wedge dx^3, dx^1 \wedge dx^3 \rangle_{\wedge} \\
 &\quad + \alpha_3 \beta_3 \langle dx^2 \wedge dx^3, dx^2 \wedge dx^3 \rangle_{\wedge} dx^1 \wedge dx^2 \wedge dx^3
 \end{aligned}$$

It follows that $\beta_3 = \alpha_1 \beta_1 + \alpha_2 \beta_2 + \alpha_3 \beta_3$ and since β was arbitrary we can plugin
 different values for $\beta_1, \beta_2, \beta_3$: $\beta_1 = 1, \beta_2 = 0, \beta_3 = 0$ lead to $\alpha_1 = 0$,
 $\beta_1 = 0, \beta_2 = 1, \beta_3 = 0$ lead to $\alpha_2 = 0$, $\beta_1 = 0, \beta_2 = 0, \beta_3 = 1$ lead to $\alpha_3 = 1$ and from
 this we get



The Hodge star operator is a linear operator on the exterior algebra $\Lambda(\mathbb{R}^n)$, mapping k -forms to $(n - k)$ -forms

$$\star : \Lambda^k(\mathbb{R}^n) \rightarrow \Lambda^{(n-k)}(\mathbb{R}^n), 0 \leq k \leq n$$

It operates on the **orthogonal basis elements** of $\Lambda_p^k(\mathbb{R}^n)$ as follows: For any increasing sequence $(i_1, \dots, i_k)$ of elements $i_p \in \{1, \dots, n\}$, if $(j_1, \dots, j_{n-k})$ is the increasing sequence of elements $j_q \in \{1, \dots, n\}$ such that

$$\{i_1, \dots, i_k\} \sqcup \{j_1, \dots, j_{n-k}\} = \{1, \dots, n\},$$

then

$$\star(dx^{i_1} \wedge \dots \wedge dx^{i_k}) = \text{sign}(i_1, \dots, i_k, j_1, \dots, j_{n-k}) dx^{j_1} \wedge \dots \wedge dx^{j_{n-k}}.$$



In particular, for $k = 0$ and $k = n$ we have

$$\star(1) = dx^1 \wedge \cdots \wedge dx^n$$

$$\star(dx^1 \wedge \cdots \wedge dx^n) = 1.$$

If $n = 3$ we have (for the derivation see example above)

$$\star(dx^1) = dx^2 \wedge dx^3$$

$$\star(dx^2) = -dx^1 \wedge dx^3$$

$$\star(dx^3) = dx^1 \wedge dx^2$$

$$\star(dx^1 \wedge dx^2) = dx^3$$

$$\star(dx^1 \wedge dx^3) = -dx^2$$

$$\star(dx^2 \wedge dx^3) = dx^1.$$



Hodge Star



- ▶ The Hodge star is an orthogonal complement for k -forms.
- ▶ But with orientation and magnitude.
- ▶ When α is a k -form on an n -dimensional Manifold, then $\star\alpha$ is a $(n - k)$ -form.
- ▶ In particular, $\alpha \wedge \star\alpha$ is a n -form with positive orientation.

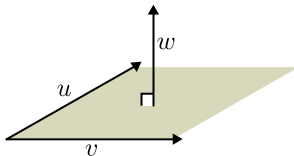
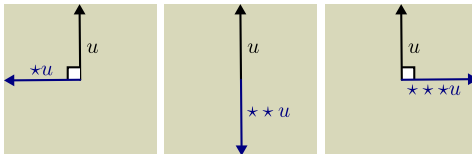


Figure: In 3D is $w = \star(u \wedge v)$ is like an orthogonal complement for the 2-vector $u \wedge v$.

In 2D, the hodge star can be seen as a rotation⁵. One can also see the sign from the alternating property of the wedge product, which leads to $\star\star u = -u$.



⁵C. Keenan. CS 15-458/858: Discrete Differential Geometry. Mar. 2020. URL: <https://brickisland.net/DDGSpring2020/>.



If we consider a general k form α spanned by $\{\sigma^{i_1}, \dots, \sigma^{i_k}\}$, i.e. $\alpha = \sum_i \alpha_i \sigma^i$, then

$$\star \alpha = \sqrt{|G|} \sum_i \alpha_i^* (\star \sigma^i),$$

where $|G|$ represents the primal metric tensor determinant and the effect on each covariant coefficient α_i is to convert it to its contravariant version $\hat{\alpha}^i$ via the dual metric tensor G^{-1} (i.e., the $\sharp$ -Operator).

Example (Diagonal Metric Tensor)

If $\beta = \beta_1 d\bar{x}^1 + \beta_2 d\bar{x}^2 + \beta_3 d\bar{x}^3$, and the metric tensor

$$G = \begin{pmatrix} \left\langle \frac{\partial}{\partial \bar{x}^1}, \frac{\partial}{\partial \bar{x}^1} \right\rangle & \left\langle \frac{\partial}{\partial \bar{x}^1}, \frac{\partial}{\partial \bar{x}^2} \right\rangle & \left\langle \frac{\partial}{\partial \bar{x}^1}, \frac{\partial}{\partial \bar{x}^3} \right\rangle \\ \left\langle \frac{\partial}{\partial \bar{x}^2}, \frac{\partial}{\partial \bar{x}^1} \right\rangle & \left\langle \frac{\partial}{\partial \bar{x}^2}, \frac{\partial}{\partial \bar{x}^2} \right\rangle & \left\langle \frac{\partial}{\partial \bar{x}^2}, \frac{\partial}{\partial \bar{x}^3} \right\rangle \\ \left\langle \frac{\partial}{\partial \bar{x}^3}, \frac{\partial}{\partial \bar{x}^1} \right\rangle & \left\langle \frac{\partial}{\partial \bar{x}^3}, \frac{\partial}{\partial \bar{x}^2} \right\rangle & \left\langle \frac{\partial}{\partial \bar{x}^3}, \frac{\partial}{\partial \bar{x}^3} \right\rangle \end{pmatrix} = \begin{pmatrix} g_{11} & 0 & 0 \\ 0 & g_{22} & 0 \\ 0 & 0 & g_{33} \end{pmatrix} \text{ is simplified such}$$

that only the non-diagonal elements are non-zero and we denote the entries of the dual metric tensor G^{-1} by $g^{ii} = \frac{1}{g_{ii}}$ then we get⁶

$$\star \beta = \frac{1}{\sqrt{g^{11} g^{22} g^{33}}} (b_1 g^{11} d\bar{x}^2 \wedge d\bar{x}^3 + b_2 g^{22} d\bar{x}^3 \wedge d\bar{x}^1 + b_3 g^{33} d\bar{x}^1 \wedge d\bar{x}^2)$$

⁶L. J. Grady and J. R. Polimeni. *Discrete calculus: Applied analysis on graphs for computational science*. Springer Science & Business Media, 2010.



To derive this formula we represent the one-forms in the orthonormal basis

$$d\bar{x}^i = \frac{1}{\sqrt{g_{ii}}} dx^i \text{ so that } \langle d\bar{x}^i, d\bar{x}^i \rangle = \frac{1}{g_{ii}} \langle dx^i, dx^i \rangle.$$

Then we get

$$\begin{aligned} \star d\bar{x}^1 &= \frac{1}{\sqrt{g_{11}}} \star dx^1 = \frac{1}{\sqrt{g_{11}}} dx^2 \wedge dx^3 = \\ &= \frac{\sqrt{g_{22}g_{33}}}{\sqrt{g_{11}}} d\bar{x}^2 \wedge d\bar{x}^3 = \frac{\sqrt{g_{11}g_{22}g_{33}}}{g_{11}} d\bar{x}^2 \wedge d\bar{x}^3 \\ &= \frac{g^{11}}{\sqrt{g^{11}g^{22}g^{33}}} d\bar{x}^2 \wedge d\bar{x}^3 \end{aligned}$$

In a similar way the get

$$\begin{aligned} \star d\bar{x}^2 &= \frac{g^{22}}{\sqrt{g^{11}g^{22}g^{33}}} d\bar{x}^3 \wedge d\bar{x}^1 \\ \star d\bar{x}^3 &= \frac{g^{33}}{\sqrt{g^{11}g^{22}g^{33}}} d\bar{x}^1 \wedge d\bar{x}^2 \end{aligned}$$



$\star(1)$ in terms of a basis

Let $T_p^*(\mathbb{R}^n)$ be oriented and $(\sigma^1, \dots, \sigma^n)$ be any basis of $T_p^*(\mathbb{R}^n)$. Then we have

$$\star(1) = \frac{1}{\sqrt{\det(\langle \sigma^i, \sigma^j \rangle_{G^{-1}})}} \sigma^1 \wedge \dots \wedge \sigma^n.$$

Proof⁷

If $(dx^1, \dots, dx^n)$ is an orthonormal basis of $T_p^*(\mathbb{R}^n)$ and $(\sigma^1, \dots, \sigma^n)$ is any other basis of $T_p^*(\mathbb{R}^n)$, then $\langle \sigma^1 \wedge \dots \wedge \sigma^n, \sigma^1 \wedge \dots \wedge \sigma^n \rangle_{ij} \wedge = \det(\langle \sigma^i, \sigma^j \rangle)$ and since $\sigma^1 \wedge \dots \wedge \sigma^n = \det(A) dx^1 \wedge \dots \wedge dx^n$, where A is the matrix expressing the σ^j in terms of the dx^i , we have

$$\langle \sigma^1 \wedge \dots \wedge \sigma^n, \sigma^1 \wedge \dots \wedge \sigma^n \rangle_{ij} \wedge = \det(A)^2 \langle dx^1 \wedge \dots \wedge dx^n, dx^1 \wedge \dots \wedge dx^n \rangle_{\wedge} = \det(A)^2.$$

Therefore, $\det(A) = \sqrt{\det(\langle \sigma^i, \sigma^j \rangle)}$, and

$$\sigma^1 \wedge \dots \wedge \sigma^n = \sqrt{\det(\langle \sigma^i, \sigma^j \rangle)} dx^1 \wedge \dots \wedge dx^n.$$

⁷J. Gallier and J. Quaintance. *Differential Geometry and Lie Groups: A Second Course*. en. Vol. 13. Geometry and Computing. Cham: Springer International Publishing, 2020. URL: <http://link.springer.com/10.1007/978-3-030-46047-1> (visited on 10/27/2021).



The Laplacian is one of the central differential operators:

- ▶ fundamental physical laws (any diffusion process and all forms of wave propagation, including the Schroedinger equation);
- ▶ its eigenvalues capture almost everything there is to know about a given piece of geometry
- ▶ applications in geometry processing include surface parameterization, surface smoothing, vector field design and decomposition, distance computation, fluid simulation, etc.

Laplacian operator

$$\Delta := \text{div} \circ \text{grad}$$

Laplacian operator in differential form

$$\Delta = \star d \star d$$

Proof:

- ▶ The gradient of ϕ is given by $(d\phi)^\sharp$. b converts this gradient the 1-form $d\phi$.
- ▶ The hodge star $\star$ converts the 1-form into the 2-form $\star d\phi$
- ▶ differentiating the 2-form $\star d\phi$ delivers $d \star d\phi$
- ▶ The hodge star $\star$ turns this 3-form into a 0-form $\star d \star d\phi$



The Laplacian operator



Note, that for a 1-form α in $\mathbb{R}^3$, the divergence operator is given by

$$\operatorname{div} := \star d \star$$

This result can be generalized to k -forms :

The Laplace-de Rham operator

$$\Delta : \Lambda^k \rightarrow \Lambda^k; \Delta := \star d \star d + d \star d \star = (d + d^\star)^2$$

Codifferential Operator

The operator $\delta := (-1)^{n(k+1)+1} \star d \star : \Lambda^k \rightarrow \Lambda^{k-1}$, is called **codifferential** operator.

Using δ we get

$$\Delta = \delta d + d \delta$$

Note, the Laplace-de Rham operator on 0-forms is identical up to sign to the Laplacian operator on scalar functions. If we apply the term $d \star d \star$ to a 0-form ϕ , $\star \phi$ is an n -form and $d \star \phi$ is an $(n+1)$ -form which does not exist in the n -dimensional space.



Let $f : M \rightarrow N$ be a mapping between two manifolds, where $f = (f_1, f_2, \dots, f_n)$ and each component function f_i is differentiable at a point $p \in M$. If the coordinates of the manifold M are given by $x_1, \dots, x_n$, then the derivative of the map f at the point $p \in M$ is given by the Jacobian Matrix

$$T_p f = \begin{pmatrix} \left. \frac{\partial f_1}{\partial x^1} \right|_p & \left. \frac{\partial f_1}{\partial x^2} \right|_p & \cdots & \left. \frac{\partial f_1}{\partial x^n} \right|_p \\ \left. \frac{\partial f_2}{\partial x^1} \right|_p & \left. \frac{\partial f_2}{\partial x^2} \right|_p & \cdots & \left. \frac{\partial f_2}{\partial x^n} \right|_p \\ \vdots & \vdots & \ddots & \vdots \\ \left. \frac{\partial f_n}{\partial x^1} \right|_p & \left. \frac{\partial f_n}{\partial x^2} \right|_p & \cdots & \left. \frac{\partial f_n}{\partial x^n} \right|_p \end{pmatrix}$$

Suppose we have some vector $v \in T_p M$. $T_p f \cdot v_p$ gives the linear approximation of the change in f as we move along v_p , which is another vector in $T_{f(p)} N$.

Since we can find $T_p f : T_p M \rightarrow T_{f(p)} N$ at each point $p \in M$, we end up with a mapping Tf from the tangent bundle of M to the tangent bundle of N , i.e.

$f : M \rightarrow N$ induces the map

$$Tf : TM \rightarrow TN.$$

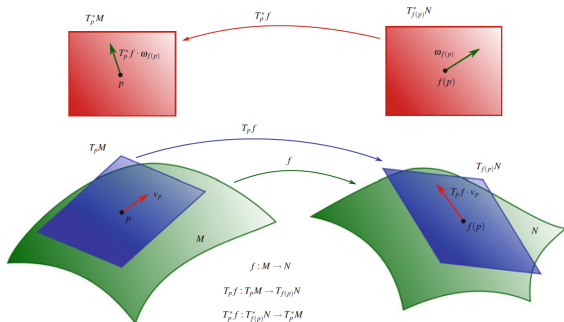
This mapping is called the **push-forward mapping** and is often denoted by Df .



Pull backs



$$\begin{aligned}
 M &\xrightarrow{f} N \\
 T_p(M) &\xrightarrow{T_p f} T_{f(p)} N \\
 N &\xrightarrow{\omega} \mathbb{R} \\
 \Lambda_p^k(M) &\xleftarrow{T_p^*} \Lambda_{f(p)}^k(N) \\
 M &\xrightarrow{T^* f \cdot \omega} \mathbb{R}.
 \end{aligned}$$



Definition (Pull back)

Given $f : M \rightarrow N$ as above, suppose that we have a differential k -form ω on N . We define the differential k -form $T^* f \cdot \omega$ as the **pull-back** of ω to M where

$$(T^* f \cdot \omega)(v_1, \dots, v_n) := \omega(T f \cdot v_1, \dots, T f \cdot v_k).$$

The pull-back map is sometimes called cotangent map and is often denoted by f^* . Notice that when it comes to the pull-back map, it is indexed by the base point in the image and not the domain.



Pull-back identities

1. $T^*\Phi \cdot (c\alpha + \beta) = cT^*\Phi \cdot \alpha + T^*\Phi \cdot \beta, c \in \mathbb{R}$
2. $T^*\Phi \cdot (\alpha \wedge \beta) = T^*\Phi \cdot \alpha \wedge T^*\Phi \cdot \beta,$
3. $T^*\Phi \cdot d\alpha = d(T^*\Phi \cdot \alpha).$

In general there is no single formula that one can use to find the pull-back of a differential form.

However, in the case of the **pull-back of a volume form** using a change in basis given by $\Phi : \mathbb{R}_{x^1, \dots, x^n}^n \rightarrow \mathbb{R}_{\Phi^1, \dots, \Phi^n}^n$ we get

$$T^*\Phi \cdot (\Phi^1 \wedge \dots \wedge \Phi^n) = \begin{vmatrix} \frac{\partial \Phi^1}{\partial x^1} & \dots & \frac{\partial \Phi^1}{\partial x^n} \\ \vdots & \ddots & \vdots \\ \frac{\partial \Phi^n}{\partial x^1} & \dots & \frac{\partial \Phi^n}{\partial x^n} \end{vmatrix} dx^1 \wedge \dots \wedge dx^n.$$

Proof: See e.g. Fortney, 2018

Definition

Let $\omega = dx^1 \wedge dx^2$ be a differential 2-form on the domain R , $f : R \rightarrow \mathbb{R}$ a function and $V_{i,j}^1, V_{i,j}^2$ be basis vectors of the tangent space $T_{(x_i, y_j)} \mathbb{R}^2$. The integral of f with respect to $dx^1 \wedge dx^2$ is given by

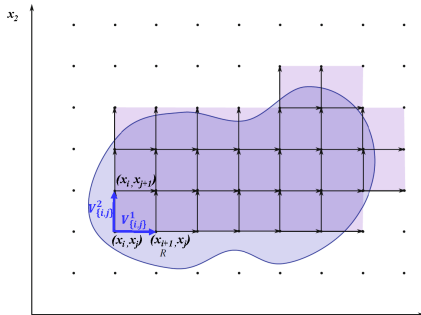
$$\int_R f dx^1 \wedge dx^2 := \lim_{m,n \rightarrow \infty} \sum_{i=0}^m \sum_{j=0}^n f(x_i, y_j) dx^1 \wedge dx^2 (V_{i,j}^1, V_{i,j}^2).$$

If the calculation of the **area form** results in

$$dx^1 \wedge dx^2 (V_{i,j}^1, V_{i,j}^2) = \Delta x_i \Delta y_j$$

we can evaluate the integral by evaluating the standard Riemann integral:

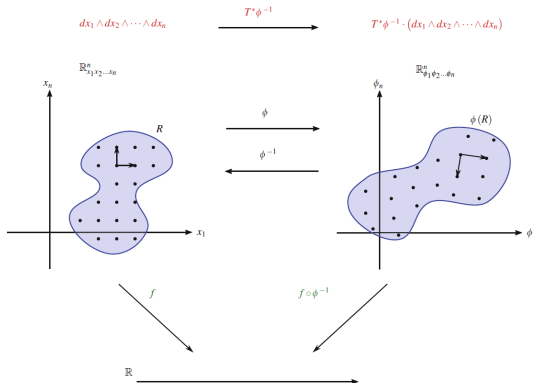
$$\int_R f dx^1 \wedge dx^2 = \int_R f(x, y) dx dy$$



When we change variables we have to change three things in our integral

- ▶ the region (blue),
- ▶ the function (green),
- ▶ the volume form (red)

The vectors on the left are pushed forward by $T\Phi$ to the vectors on the right.



Change of coordinates formula

Let $\Phi : \mathbb{R}_{x_1, x_2, \dots, x_n}^n \rightarrow \mathbb{R}_{\Phi_1, \Phi_2, \dots, \Phi_n}^n$. Then we get the following formula for the integral:

$$\int_R f(x_1, \dots, x_n) dx^1 \wedge \dots \wedge dx^n = \int_{\Phi(R)} f \circ \Phi^{-1}(\Phi_1, \dots, \Phi_n) T^*\Phi^{-1}(dx^1 \wedge \dots \wedge dx^n).$$



1. So far we considered flat spaces like $\mathbb{R}^n$.
2. We are interested in differential forms and vectors in the general setting of differentiable manifolds as they are expressed in a manner that is not dependent on a particular coordinate system.
3. A fundamental distinction between vectors and forms is how they react to a change of coordinates to preserve their coordinate invariance.
4. Elements of the tangent space $T\mathcal{M}^n$ behave as **contravariant tensors**, i.e. operate in the *opposite* direction with a change of coordinates and elements of the cotangent space $T^*\mathcal{M}^n$ behave as **covariant tensors**, i.e. vary in the same direction as the coordinates.

8



⁸Consider one-dimensional coordinate system defined in the unit of centimeter with a position vector at the origin extending to the location at 500cm . If the measurement unit changes from centimeters to meters, the basis vectors will become longer and the position vector will get smaller.



Integration of Differential Forms on parametrized surfaces

As long as we have a parameterization of the manifold $\Sigma_k \subset \mathbb{R}^n$, $k < n$, that is, an invertible mapping $\Phi^{-1} : U \subset \mathbb{R}^k \rightarrow \Sigma_k \subset \mathbb{R}^n$, we can integrate an k -form on Σ_k by pulling the k -form back to $U \subset \mathbb{R}^k$ using $T^*\Phi^{-1}$ and integrating there

$$\int_{\Sigma_k} \alpha = \int_{\Phi^{-1}(\Sigma_k)} T^*\Phi^{-1} \cdot \alpha.$$

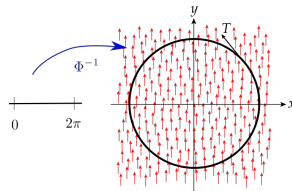


Example (1)

Let $\omega = dy$ be a 1-form in the plane, which we will integrate over the unit circle S^1 parameterized by $\Phi^{-1} : [0, 2\pi) \rightarrow \mathbb{R}^2$; $\Phi^{-1}(t) = (\cos(t), \sin(t))^t$.

We first note, that $T\Phi^{-1} : T_p\mathbb{R} \rightarrow T_{\Phi^{-1}(p)}(\mathbb{R}^2)$ is given by

$$\begin{aligned} T_p(\mathbb{R}) \ni [1] &\mapsto T_p\Phi^{-1} \cdot [1] = \begin{bmatrix} -\sin(t) \\ \cos(t) \end{bmatrix} [1] = \begin{bmatrix} -\sin(t) \\ \cos(t) \end{bmatrix} \\ &= \left(-\sin(t) \frac{\partial}{\partial x} + \cos(t) \frac{\partial}{\partial y} \right) \in T_{\Phi^{-1}(p)}(\mathbb{R}^2) \end{aligned}$$





Further, the pull-back map of $\omega_{\Phi^{-1}(p)} = dy$ is a one form on $T_p(\mathbb{R})$, i.e. $T^*\Phi^{-1}(dy) = f(t)dt$ for some function $f(t)$ and $dt \in \Lambda^1(\mathbb{R})$.

Taking the unit vector $[1] \in T_p(\mathbb{R})$ and noting that $dt([1]) = 1$ we get

$$\begin{aligned} f(t) &= f(t)dt([1]) = (T^*\Phi^{-1} \cdot dy)([1]) = dy(T\Phi^{-1} \cdot [1]) \\ &= dy\left(-\sin(t)\frac{\partial}{\partial x} + \cos(t)\frac{\partial}{\partial y}\right) = \cos(t) \end{aligned}$$

Therefore, the integral can be computed as

$$\begin{aligned} \int_{S^1} dy &= \int_{\Phi^{-1}(S^1)} T^*\Phi^{-1}(dy) \\ &= \int_0^{2\pi} \cos(t)dt \\ &= 0. \end{aligned}$$



Example (2)

We will find the integral of the one-form $x_2^2 dx^1 + x_1 dx^2$ along $C := \{\Phi^{-1}(t) = (x_1(t), x_2(t)) | x_1 = 4 - t^2, x_2 = t, -3 \leq t \leq 2\}$. We first note, that

$$T\Phi^{-1} : T_p R \rightarrow T_{\Phi^{-1}(p)}(\mathbb{R}^2),$$

$$T_p(\mathbb{R}) \ni [1] \mapsto T_p\Phi^{-1} \cdot [1] = \begin{bmatrix} -2t \\ 1 \end{bmatrix} [1] = \begin{bmatrix} -2t \\ 1 \end{bmatrix} \in T_{\Phi^{-1}(p)}(\mathbb{R}^2)$$

Further, $T^*\Phi^{-1}(x_2^2 dx^1 + x_1 dx^2) = f(t)dt$ for some function $f(t)$. We have

$$\begin{aligned} f(t) &= f(t)dt([1]) \\ &= (T^*\Phi^{-1} \cdot (x_2^2 dx^1 + x_1 dx^2))([1]) \\ &= (x_2^2 dx^1 + x_1 dx^2)(T\Phi^{-1} \cdot [1]) = (x_2^2 dx^1 + x_1 dx^2)\left(\begin{bmatrix} -2t \\ 1 \end{bmatrix}\right) \\ &= x_2^2 dx_1\left(\begin{bmatrix} -2t \\ 1 \end{bmatrix}\right) + x_1 dx_2\left(\begin{bmatrix} -2t \\ 1 \end{bmatrix}\right) \\ &= x_2^2(-2t) + x_1(1) = (t^2)(-2t) + (4 - t^2)(1) = -2t^3 - t^2 + 4 \end{aligned}$$

Therefore,

$$\int_C x_2^2 dx_1 + x_1 dx_2 = \int_{[-3,2]} (-2t^3 - t^2 + 4) dt.$$



Example (3) Volume of upper half of a unit sphere in $\mathbb{R}^3$

Integrate the 2-form $\omega = \sqrt{1 - x_1^2 - x_2^2} dx_1 \wedge dx_2$ over $U = \{x_1^2 + x_2^2 \leq 1\}$.

To solve this problem we use polar coordinates $\Phi^{-1}(r, \theta) = (r\cos(\theta), r\sin(\theta))$, where $(r, \theta) \in \Phi(U) = [0, 1] \times [0, 2\pi]$ and apply the change of coordinates formula.

$$\begin{aligned}
 \int_U \sqrt{1 - x_1^2 - x_2^2} dx_1 \wedge dx_2 &= \int_{\Phi(U)} \underbrace{\sqrt{1 - x_1^2 - x_2^2} \circ \Phi^{-1}(r, \theta)}_{\sqrt{1 - r^2 \cos^2(\theta) - r^2 \sin^2(\theta)}} \underbrace{T^* \Phi^{-1} \cdot (dx_1 \wedge dx_2)}_{\text{pull-back volume form}} \\
 &= \int_{\Phi(U)} \sqrt{1 - r^2} \left| \begin{array}{cc} \frac{\partial x_1}{\partial r} & \frac{\partial x_1}{\partial \theta} \\ \frac{\partial x_2}{\partial r} & \frac{\partial x_2}{\partial \theta} \end{array} \right| d\theta \wedge dr \\
 &= \int_{\Phi(U)} \sqrt{1 - r^2} \left| \begin{array}{cc} -r\sin(\theta) & \cos(\theta) \\ r\cos(\theta) & \sin(\theta) \end{array} \right| d\theta \wedge dr \\
 &= \int_{\Phi(U)} \sqrt{1 - r^2} (-r) d\theta \wedge dr \\
 &= \int_0^{2\pi} \int_0^1 -r \sqrt{1 - r^2} d\theta dr = -2\pi \frac{2}{2 \cdot 3} [(1 - u)^{\frac{3}{2}}]_0^1 = \frac{2\pi}{3}.
 \end{aligned}$$



Example (4) Surface of upper half of a unit sphere in $\mathbb{R}^3$

Let Ω denote the top half of the unit sphere in $\mathbb{R}^3$ and $\omega = \frac{1}{\sqrt{1-x_3^2}} dx^1 \wedge dx^2$ be a differential 2-form on $\mathbb{R}^3$. Calculate $\int_{\Omega} \omega$.

We employ the parameterization

$$\Phi^{-1}(r, t) = (x_1(r, t), x_2(r, t), x_3(r, t)) = (r \cos(t), r \sin(t), \sqrt{1-r^2}),$$

where $(r, t) \in R = [0, 1] \times [0, 2\pi]$ and $\Omega = \Phi^{-1}R$.

The pull-back map of $T^*\Phi^{-1} \cdot \left(\frac{1}{\sqrt{1-x_3^2}} dx^1 \wedge dx^2 \right)$ is a two-form on $\mathbb{R}^2$, i.e. it is of the form $g(r, t)dr \wedge dt$. To find $g(r, t)$ we notice that

$$g(r, t)dr \wedge dt \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) = g(r, t) \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = g(r, t)$$



Using this observation we compute

$$\begin{aligned}
 g(r, t) &= T^* \Phi^{-1} \left(\frac{1}{\sqrt{1-x_3^2}} dx^1 \wedge dx^2 \right) \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\
 &= \left(\frac{1}{\sqrt{1-x_3^2}} dx^1 \wedge dx^2 \right) \left(T\Phi^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, T\Phi^{-1} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\
 &= \frac{1}{\sqrt{1-x_3^2}} dx^1 \wedge dx^2 \left(\begin{pmatrix} \cos(\theta) \\ \sin(\theta) \\ \frac{-r}{\sqrt{1-r^2}} \end{pmatrix}, \begin{pmatrix} -r\sin(\theta) \\ r\cos(\theta) \\ 0 \end{pmatrix} \right) \\
 &= \frac{1}{\sqrt{1-x_3^2}} \begin{vmatrix} \cos(\theta) & -r\sin(\theta) \\ \sin(\theta) & r\cos(\theta) \end{vmatrix} = \frac{1}{\sqrt{1-x_3^2}} r = \frac{1}{r} r = 1
 \end{aligned}$$

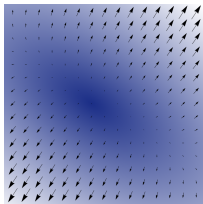
Finally, we get

$$\begin{aligned}
 \int_{\Phi^{-1}(R)} \omega &= \int_R T^* \Phi^{-1} \cdot (x_3^2 dx^1 \wedge dx^2) = \int_R 1 dr \wedge dt \\
 &= \int_0^{2\pi} \int_0^1 dr dt = 2\pi.
 \end{aligned}$$

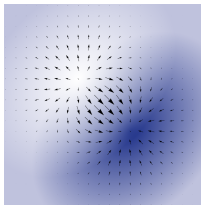
Vector Derivatives in Coordinates

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a scalar function and $\mathbf{F} = P \frac{\partial}{\partial x^1} + Q \frac{\partial}{\partial x^2} + R \frac{\partial}{\partial x^3}$ be a vector field, where $P, Q, R : \mathbb{R}^3 \rightarrow \mathbb{R}$ are coordinate functions that vary over the domain, and $\frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3}$ are the standard basis vector fields.

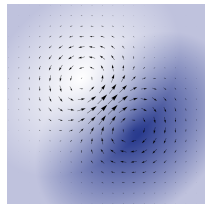
grad: ∇f



div: $\nabla \cdot \mathbf{F}$



curl: $\nabla \times \mathbf{F}$



$$\nabla f = \frac{\partial f}{\partial x^1} \frac{\partial}{\partial x^1} + \frac{\partial f}{\partial x^2} \frac{\partial}{\partial x^2} + \frac{\partial f}{\partial x^3} \frac{\partial}{\partial x^3}$$

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x^1} + \frac{\partial Q}{\partial x^2} + \frac{\partial R}{\partial x^3} \quad \nabla \times \mathbf{F} = \left(\frac{\partial R}{\partial x^2} - \frac{\partial Q}{\partial x^3} \right) \frac{\partial}{\partial x^1} + \left(\frac{\partial P}{\partial x^3} - \frac{\partial R}{\partial x^1} \right) \frac{\partial}{\partial x^2} + \left(\frac{\partial Q}{\partial x^1} - \frac{\partial P}{\partial x^2} \right) \frac{\partial}{\partial x^3}$$



Gradient and exterior derivative

Let $f \in C(\mathbb{R}^3)$. Then

$$df = (\nabla f)^\flat,$$

i.e. the diagram on the right commutes.

$$\begin{array}{ccc} C(\mathbb{R}^3) & \xrightarrow{\text{grad}} & T\mathbb{R}^3 \\ \downarrow id & & \downarrow \flat \\ \Lambda^0(\mathbb{R}^3) & \xrightarrow{d} & \Lambda^1(\mathbb{R}^3) \end{array}$$

Proof:

$$T(\mathbb{R}^3) \ni \nabla f = \frac{\partial f}{\partial x^1} \frac{\partial}{\partial x^1} + \frac{\partial f}{\partial x^2} \frac{\partial}{\partial x^2} + \frac{\partial f}{\partial x^3} \frac{\partial}{\partial x^3}$$

Flattening leads to

$$\begin{aligned} (\nabla f)^\flat &= \frac{\partial f}{\partial x^1} dx^1 + \frac{\partial f}{\partial x^2} dx^2 + \frac{\partial f}{\partial x^3} dx^3 \\ &= df \end{aligned}$$

□



Curl and exterior derivative

Let $T(\mathbb{R}^3) \ni \mathbf{F} = P \frac{\partial}{\partial x^1} + Q \frac{\partial}{\partial x^2} + R \frac{\partial}{\partial x^3}$ be a vector field. Then we get

$$d(\mathbf{F}^\flat) = \star((\text{curl } \mathbf{F})^\flat) = (\star \circ \flat)(\text{curl } \mathbf{F}),$$

i.e. the diagram on the right commutes.

$$\begin{array}{ccc} T(\mathbb{R}^3) & \xrightarrow{\text{curl}} & T\mathbb{R}^3 \\ \downarrow \flat & & \downarrow \star \circ \flat \\ \Lambda^1(\mathbb{R}^3) & \xrightarrow{d} & \Lambda^2(\mathbb{R}^3) \end{array}$$

Proof:

$$\text{curl } \mathbf{F} = \left(\frac{\partial R}{\partial x^2} - \frac{\partial Q}{\partial x^3} \right) \frac{\partial}{\partial x^1} + \left(\frac{\partial P}{\partial x^3} - \frac{\partial R}{\partial x^1} \right) \frac{\partial}{\partial x^2} + \left(\frac{\partial Q}{\partial x^1} - \frac{\partial P}{\partial x^2} \right) \frac{\partial}{\partial x^3}$$

So we get

$$\begin{aligned} \star(\mathbf{F}^\flat) &= \star \left(\left(\frac{\partial R}{\partial x^2} - \frac{\partial Q}{\partial x^3} \right) dx^1 + \left(\frac{\partial P}{\partial x^3} - \frac{\partial R}{\partial x^1} \right) dx^2 + \left(\frac{\partial Q}{\partial x^1} - \frac{\partial P}{\partial x^2} \right) dx^3 \right) \\ &= \left(\frac{\partial R}{\partial x^2} - \frac{\partial Q}{\partial x^3} \right) dx^2 \wedge dx^3 + \left(\frac{\partial P}{\partial x^3} - \frac{\partial R}{\partial x^1} \right) dx^3 \wedge dx^1 \\ &\quad + \left(\frac{\partial Q}{\partial x^1} - \frac{\partial P}{\partial x^2} \right) dx^1 \wedge dx^2. \end{aligned}$$



We already computed

$$(b) \mathbf{F} \equiv \mathbf{F}^b = Pdx^1 + Qdx^2 + Rdx^3.$$

Taking the exterior derivative

$$\begin{aligned} d(\mathbf{F}^b) &= d(Pdx^1 + Qdx^2 + Rdx^3) \\ &= dP \wedge dx^1 + dQ \wedge dx^2 + dR \wedge dx^3 \\ &= \left(\frac{\partial P}{\partial x^1} dx^1 + \frac{\partial P}{\partial x^2} dx^2 + \frac{\partial P}{\partial x^3} dx^3 \right) \wedge dx^1 + \left(\frac{\partial Q}{\partial x^1} dx^1 + \frac{\partial Q}{\partial x^2} dx^2 + \frac{\partial Q}{\partial x^3} dx^3 \right) \wedge dx^2 \\ &\quad + \left(\frac{\partial R}{\partial x^1} dx^1 + \frac{\partial R}{\partial x^2} dx^2 + \frac{\partial R}{\partial x^3} dx^3 \right) \wedge dx^3 \\ &= \frac{\partial P}{\partial x^2} dx^2 \wedge dx^1 + \frac{\partial P}{\partial x^3} dx^3 \wedge dx^1 + \frac{\partial Q}{\partial x^1} dx^1 \wedge dx^2 + \frac{\partial Q}{\partial x^3} dx^3 \wedge dx^2 \\ &\quad + \frac{\partial R}{\partial x^1} dx^1 \wedge dx^3 + \frac{\partial R}{\partial x^2} dx^2 \wedge dx^3 \\ &= \left(\frac{\partial R}{\partial x^2} - \frac{\partial Q}{\partial x^3} \right) dx^2 \wedge dx^3 + \left(\frac{\partial P}{\partial x^3} - \frac{\partial R}{\partial x^1} \right) dx^3 \wedge dx^1 \\ &\quad + \left(\frac{\partial Q}{\partial x^1} - \frac{\partial P}{\partial x^2} \right) dx^1 \wedge dx^2. \end{aligned}$$

□

Divergence and exterior derivative

Let $T(\mathbb{R}^3) \ni \mathbf{F} = P \frac{\partial}{\partial x^1} + Q \frac{\partial}{\partial x^2} + R \frac{\partial}{\partial x^3}$ be a vector field. Then we get

$$d((\star \circ \flat)\mathbf{F}) = \star(\operatorname{div} \mathbf{F}),$$

i.e. the diagram on the right commutes.

The following diagram commutes:

$$\begin{array}{ccc} T(\mathbb{R}^3) & \xrightarrow{\operatorname{div}} & C\mathbb{R}^3 \\ \downarrow \star \circ \flat & & \downarrow \star \\ \Lambda^2(\mathbb{R}^3) & \xrightarrow{d} & \Lambda^3(\mathbb{R}^3) \end{array}$$

Proof: Taking the Hodge star of the divergence of $\mathbf{F}$ we get

$$\begin{aligned} \star(\operatorname{div} \mathbf{F}) &= \star\left(\frac{\partial P}{\partial x^1} + \frac{\partial Q}{\partial x^2} + \frac{\partial R}{\partial x^3}\right) \\ &= \left(\frac{\partial P}{\partial x^1} + \frac{\partial Q}{\partial x^2} + \frac{\partial R}{\partial x^3}\right) dx^1 \wedge dx^2 \wedge dx^3. \end{aligned}$$

Then we compute

$$\begin{aligned}
 (\star \circ \flat)\mathbf{F} &= \star \mathbf{F}^\flat = \star \left(\left(\frac{\partial P}{\partial x^1} + \frac{\partial Q}{\partial x^2} + \frac{\partial R}{\partial x^3} \right)^\flat \right) \\
 &= \star(Pdx^1 + Qdx^2 + Rdx^3) \\
 &= Pdx^2 \wedge dx^3 + Qdx^3 \wedge dx^1 + Rdx^1 \wedge dx^2
 \end{aligned}$$

Taking the exterior derivative we get

$$\begin{aligned}
 d((\star \circ \flat)\mathbf{F}) &= d(Pdx^2 \wedge dx^3 + Qdx^3 \wedge dx^1 + Rdx^1 \wedge dx^2) \\
 &= \frac{\partial P}{\partial x^1} dx^1 \wedge dx^2 \wedge dx^3 + \frac{\partial Q}{\partial x^2} dx^2 \wedge dx^3 \wedge dx^1 + \frac{\partial R}{\partial x^3} dx^3 \wedge dx^1 \wedge dx^2 \\
 &= \left(\frac{\partial P}{\partial x^1} + \frac{\partial Q}{\partial x^2} + \frac{\partial R}{\partial x^3} \right) dx^1 \wedge dx^2 \wedge dx^3
 \end{aligned}$$

□



Relationship between the three vector calculus operators and differential forms

The following commuting diagram summarizes the relationship between vector calculus operations and the exterior derivative:

$$\begin{array}{ccccccc}
 C(\mathbb{R}^3) & \xrightarrow{\text{grad}} & T\mathbb{R}^3 & \xrightarrow{\text{curl}} & T\mathbb{R}^3 & \xrightarrow{\text{div}} & C\mathbb{R}^3 \\
 \downarrow id & & \downarrow b & & \downarrow \star \circ b & & \downarrow \star \\
 \Lambda^0(\mathbb{R}^3) & \xrightarrow{d} & \Lambda^1(\mathbb{R}^3) & \xrightarrow{d} & \Lambda^2(\mathbb{R}^3) & \xrightarrow{d} & \Lambda^3(\mathbb{R}^3)
 \end{array}$$

Based on this results the following identities from vector calculus can be written in terms of exterior derivatives:

Curl of gradient vanishes

$$\nabla \times (\nabla f) = 0 \Leftrightarrow d(df) = 0$$

Divergence of curl vanishes

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0 \Leftrightarrow d(d\alpha) = 0$$

Note, that the notions of differential forms and exterior derivatives can be generalized to both $\mathbb{R}^n$ for $n > 3$ as well as to general manifolds.

Fundamental theorem of Line Integrals

Let C be a curve in $\mathbb{R}^3$ parameterized by $c : \mathbb{R} \ni [a, b] \rightarrow \mathbb{R}^3, c(s) = (x_1(s), x_2(s), x_3(s))$ and let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a function. Then the fundamental theorem on line integrals is given by

$$\int_C (\text{grad } f) \cdot ds = f(c(b)) - f(c(a)) = \int_{\partial C} f.$$

To write the left hand side in terms of differential forms we set $\mathbf{F} := \text{grad } f$ and write it as a vector field with respect to the standard basis $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

Then we get

$$\begin{aligned} \int_C \mathbf{F} \cdot ds &\equiv \int_C \mathbf{F} \cdot c'(s) ds \\ &= \int_C (P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) \cdot \left(\frac{dx_1}{ds}\mathbf{i} + \frac{dx_2}{ds}\mathbf{j} + \frac{dx_3}{ds}\mathbf{k} \right) ds \\ &= \int_C \left(P \frac{dx_1}{ds} + Q \frac{dx_2}{ds} + R \frac{dx_3}{ds} \right) ds \\ &= \int_C \mathbf{F}^\flat \end{aligned}$$

Because of $(\text{grad } f)^\flat = df$ we get

Fundamental theorem of line integrals in differential form

$$\int_C df = \int_{\partial C} f.$$

Stokes' theorem

Let S be a parameterized surface in $\mathbb{R}^3$ with boundary ∂S parameterized by $c : \mathbb{R} \ni [a, b] \rightarrow \mathbb{R}^3, c(s) = (x_1(s), x_2(s), x_3(s))$ and let $F = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ be a vector field with continuous first order partial derivatives, then

$$\int_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} \, dS = \int_{\partial C} \mathbf{F} \cdot c'(s) \, ds.$$

In differential form the theorem can be written as follows:

Stokes' theorem in differential form

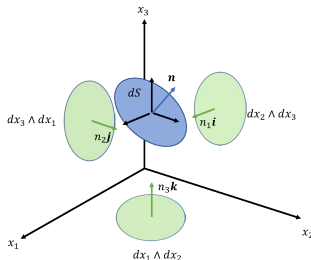
$$\int_S d(\mathbf{F}^b) = \int_{\partial S} \mathbf{F}^b$$

Vector Calculus Stokes' Theorem

Proof: As in the case of line integrals for the right hand side we get $\int_{\partial S} \mathbf{F} \cdot \mathbf{c}'(s) ds = \int_{\partial S} \mathbf{F}^b$. For the left hand side we have to show that

$$\int_S \underbrace{\text{curl } \mathbf{F}}_{\mathbf{G}} \cdot \mathbf{n} dS = \int_S \star \left(\underbrace{(\text{curl } \mathbf{F})^b}_{\mathbf{G}} \right).$$

To show that $\int_S \mathbf{G} \cdot \mathbf{n} dS = \int_S \star(\mathbf{G}^b)$, we decompose the infinitesimal area dS with normal $\mathbf{n} = n_1 \mathbf{i} + n_2 \mathbf{j} + n_3 \mathbf{k}$ into pieces in the respective planes as shown on the right.



We write the argument of the integral on the left hand side in terms of differential forms $\mathbf{G} \cdot \mathbf{n} dS = \mathbf{G} \cdot d\mathbf{S}$ and write $\mathbf{G}$ with respect to the standard basis $\mathbf{i}, \mathbf{j}, \mathbf{k}$. Then we get

$$\begin{aligned} \int_S \mathbf{G} \cdot d\mathbf{S} &= \int_S (P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) \cdot (n_1\mathbf{i} + n_2\mathbf{j} + n_3\mathbf{k}) dS \\ &= \int_S (Pn_1 + Qn_2 + Rn_3) dS = \int_S Pn_1 dS + \int_S Pn_2 dS + \int_S Rn_3 dS \\ &= \int_S P dx^2 \wedge dx^3 + Q dx^3 \wedge dx^1 + R dx^1 \wedge dx^2 = \int_S \star(\mathbf{G}^b) \quad \square \end{aligned}$$



Divergence theorem

Let $V \in \mathbb{R}^3$ be a compact subset with piecewise smooth boundary $\partial V = S$ and let $F = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ be a vector field with continuous first order partial derivatives, then

$$\int_V (\operatorname{div} \mathbf{F}) dV = \int_{\partial V} \mathbf{F} \cdot d\mathbf{S}.$$

In differential form the theorem can be written as follows:

Stokes' theorem in differential form

$$\int_V d((\star \circ \flat)\mathbf{F}) = \int_{\partial V} \star(\mathbf{F}^\flat).$$

Proof:

Notice, that the integrand of the left hand side is $\star(\operatorname{div} \mathbf{F})$, which can be rewritten as $d(\star \circ \flat)\mathbf{F}$. Furthermore, $\int_{\partial V} \mathbf{F} \cdot d\mathbf{S} = \int_{\partial V} \star(\mathbf{F}^\flat)$.



Summary: We have written all three vector calculus theorems (Fundamental Theorem for Line Integrals, Stokes' Theorem, Divergence Theorem) in the same form in differential forms notation.

Generalized Stokes' Theorem

Let α be any $n - 1$ -form on an n -dimensional domain Ω . Then we get

$$\int_{\Omega} d\alpha = \int_{\partial\Omega} \alpha.$$

Integrating a differential form over the boundary of a manifold is the same as integrating its derivative over the entire domain .



Remark

The boundary of a boundary is always zero, and thus the exterior derivative of an exterior derivative is always zero, which can be seen with the Generalized Stokes' Theorem.

Proof idea: It is a basic topological fact that the boundary of any manifold that is itself a boundary of a higher-dimensional manifold must always be empty. (e.g. the boundary of the unit ball in 3D is a closed manifold without boundary). Algebraically, for the boundary operator we get $\partial\partial = 0$. Applying the generalized Stokes' Theorem twice and using that $\partial\partial = 0$ we get

$$\int_S d(d\alpha) = \int_{\partial S} d\alpha = \int_{\partial\partial S} \alpha = 0,$$

and therefore $d(d\alpha) = 0$.



The generalized Stokes' Theorem therefore highlights how a topological property of the boundary operator relates to the topology of vector fields and motivates the following definitions:

Closed differential form

A differential form α is called **closed** if $d\alpha = 0$.

Closed differential form

A differential form α is called **exact** if there is another differential form β such that $\alpha = d\beta$.

Note, it is not always the case that a closed form is exact, e.g. the 1-form on $U \subset \mathbb{R}^2$ is closed but not exact:

$$\alpha = \frac{-x_2 dx_1 + x_1 dx_2}{x_1^2 + x_2^2}$$

Poincare Theorem

Every closed form on $\mathbb{R}^n$ is exact.⁹

⁹J. P. Fortney. *A Visual Introduction to Differential Forms and Calculus on Manifolds*. en. Cham: Springer International Publishing, 2018. URL: <http://link.springer.com/10.1007/978-3-319-96992-3> (visited on 11/03/2021).



- ▶ Book: J. Gallier and J. Quaintance. *Algebra, Topology, Differential Calculus, and Optimization Theory For Computer Science and Machine Learning*. 2019
- ▶ Book: L. J. Grady and J. R. Polimeni. *Discrete calculus: Applied analysis on graphs for computational science*. Springer Science & Business Media, 2010
- ▶ Book: D. Bachman. *A geometric approach to differential forms*. Springer Science & Business Media, 2012
- ▶ Lecture: C. Keenan. *CS 15-458/858: Discrete Differential Geometry*. Mar. 2020.
URL: <https://brickisland.net/DDGSpring2020/>



- [1] D. Bachman. *A geometric approach to differential forms*. Springer Science & Business Media, 2012.
- [2] J. P. Fortney. *A Visual Introduction to Differential Forms and Calculus on Manifolds*. en. Cham: Springer International Publishing, 2018. URL: <http://link.springer.com/10.1007/978-3-319-96992-3> (visited on 11/03/2021).
- [3] J. Gallier and J. Quaintance. *Algebra, Topology, Differential Calculus, and Optimization Theory For Computer Science and Machine Learning*. 2019.
- [4] J. Gallier and J. Quaintance. *Differential Geometry and Lie Groups: A Second Course*. en. Vol. 13. Geometry and Computing. Cham: Springer International Publishing, 2020. URL: <http://link.springer.com/10.1007/978-3-030-46047-1> (visited on 10/27/2021).
- [5] L. J. Grady and J. R. Polimeni. *Discrete calculus: Applied analysis on graphs for computational science*. Springer Science & Business Media, 2010.
- [6] C. Keenan. *CS 15-458/858: Discrete Differential Geometry*. Mar. 2020. URL: <https://brickisland.net/DDGSpring2020/>.